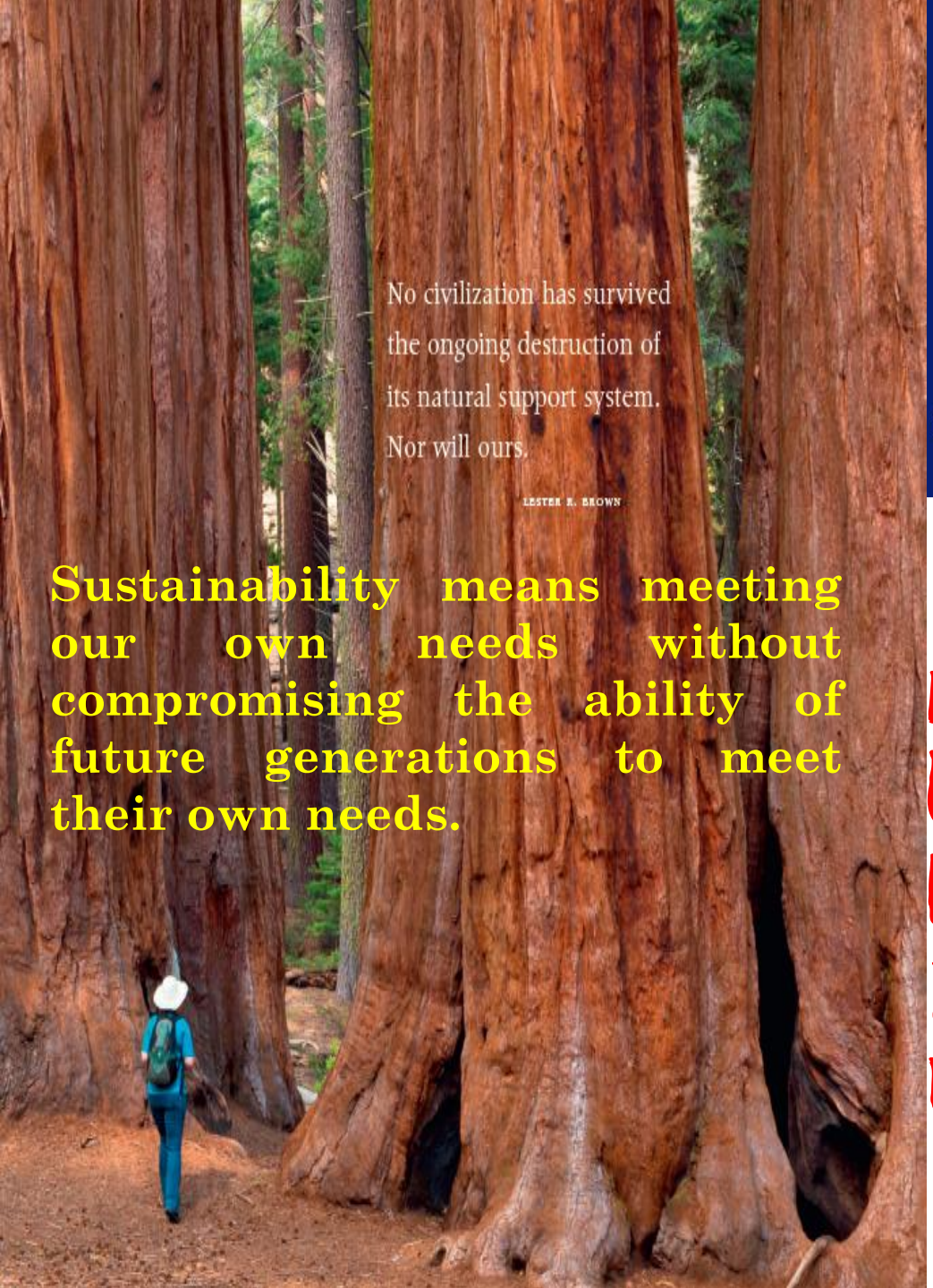


Sustainability means meeting our own needs without compromising the ability of future generations to meet their own needs.

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A Place to Learn, A Chance to Grow

Module 1:
Key Environmental Problems and Sustainability

Dr. Saurav Prasad

EVS- An Insight

- Course Code : **CHY 1006**
- Course Type : **LT**
- Credits : **2**







EVS- An Insight

Key Environmental Problems and Sustainability

Global environmental issues, socio-environmental interaction; impact of consumerism and global economy; Sustainability: Fundamentals, science behind sustainability, challenges, potential; role of technology, policies.

Ecosystem and Biodiversity

Ecosystem: functions and services, Nutrient, phosphorous, carbon, nitrogen, cycles; Effect of human activities on these cycles. Threats to biodiversity: conservation strategies, sustainable solutions.

Environmental pollution

Pollution sources, effects and mitigation, conservation. Water footprint and virtual water. Sustainable solutions. Environmental Protection Act – Air, water, forest and wild life. Interactional protocols and agreements. Resource recycling and sustainability.

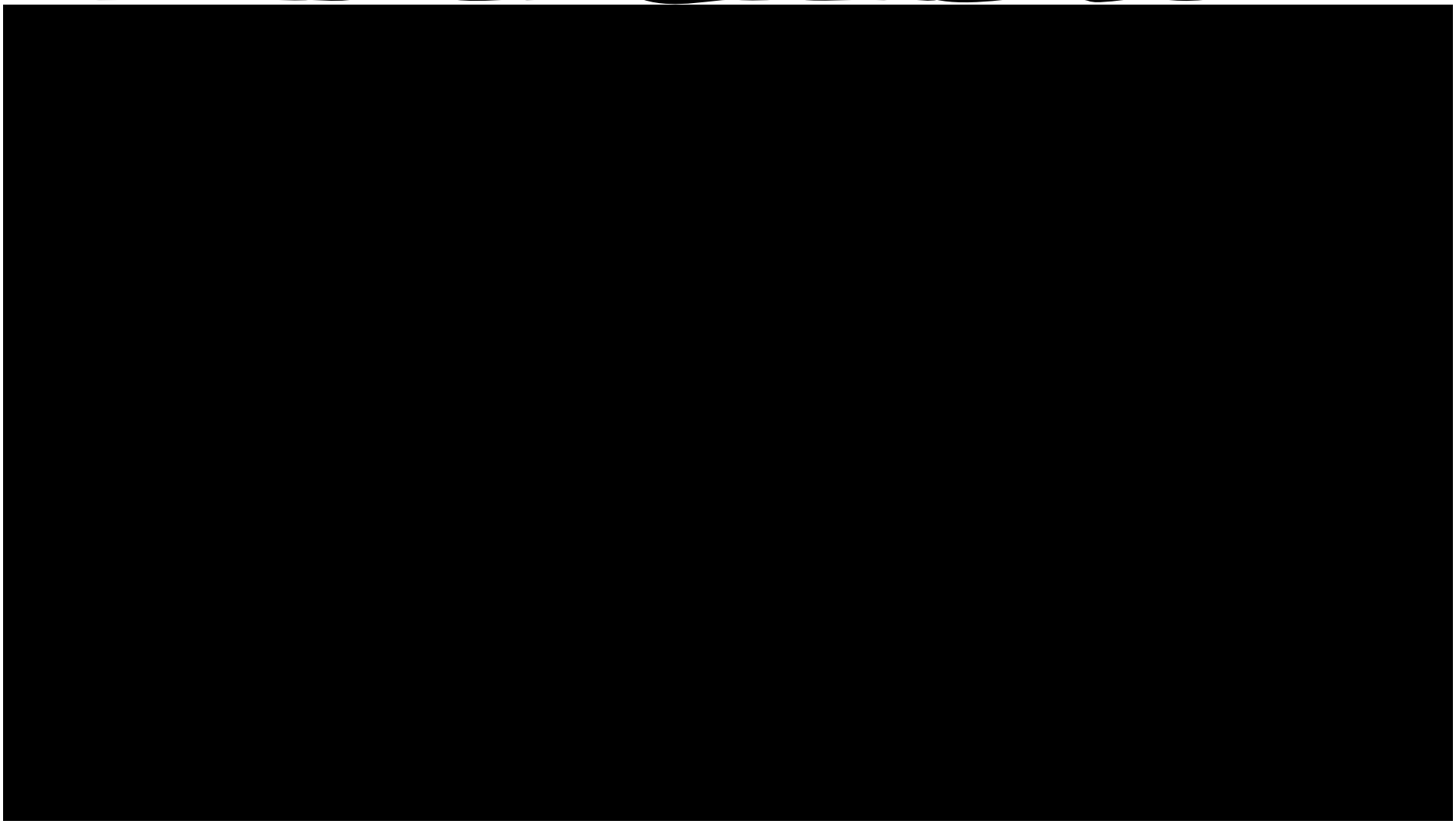
Human Population and Environment

Impact of population age structure, human population growth, Sustaining human societies: Economics, environment, policies and education.

Energy Sources and Sustainability

Conventional sources, non-conventional sources, natural resources and economics, clean coal technology, sequestration methods, energy conservation and sustainability, policies, acts, examples.

Seven Billion Dreams.
One Planet.
Consume with Care.



Environment : Total of our surroundings!!!



Consists of both:

Biotic factors (living things)

&

Abiotic factors (nonliving things)



that surround us and with which we interact



Environmental Science

- **Environment:** Everything around us. Includes living and nonliving things (air, water, and Energy) with which we interact in a complex web of relationships.

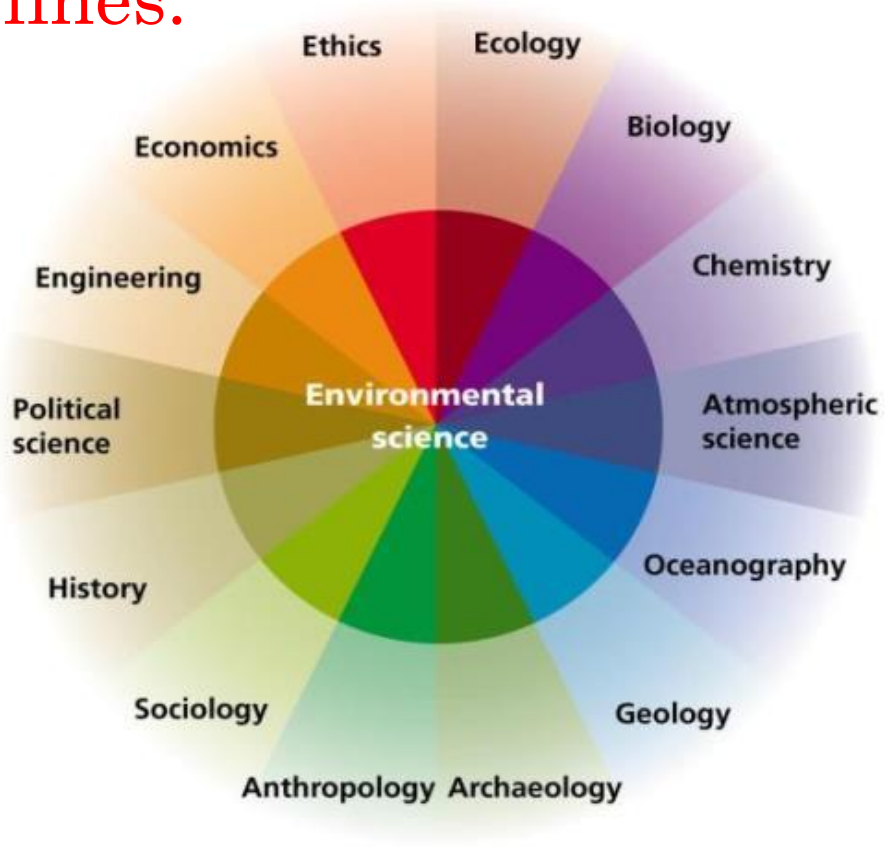
  Albert Einstein: "The environment is everything that isn't me."

- **Environmental Science:** Interdisciplinary study of how humans interact with the living and nonliving parts of their environment.

Environmental Science → Ecology → Ecosystem

Environmental Science

… is an **interdisciplinary** field, drawing on many diverse disciplines.



Environmental Science

Do we know

- **how does the natural world work?**
- **how does our environment affect us?**
- **how do we affect our environment?**

Applied goal:

- **Learn how life on earth has survived and thrived.**
- **Understand how we interact with the environment.**
- **Developing solutions to environmental problems.**



But, do we even understand the problem?













Nature's Survival Strategies:

- **Life on earth:** 3.5 Billion years; How did the incredible variety of life sustain itself for so long in the face of catastrophic changes?; meteorites, ice age, long warming periods.

- **Humans**: 200,000 years; Home sapiens: Latin for "wise man"; Really??; Species in the process of degrading its own life-support system could not be considered wise.

- **Lessons from nature or 3 principles of sustainability:**

1. Reliance on Solar Energy
2. Biodiversity
3. Chemical Cycling

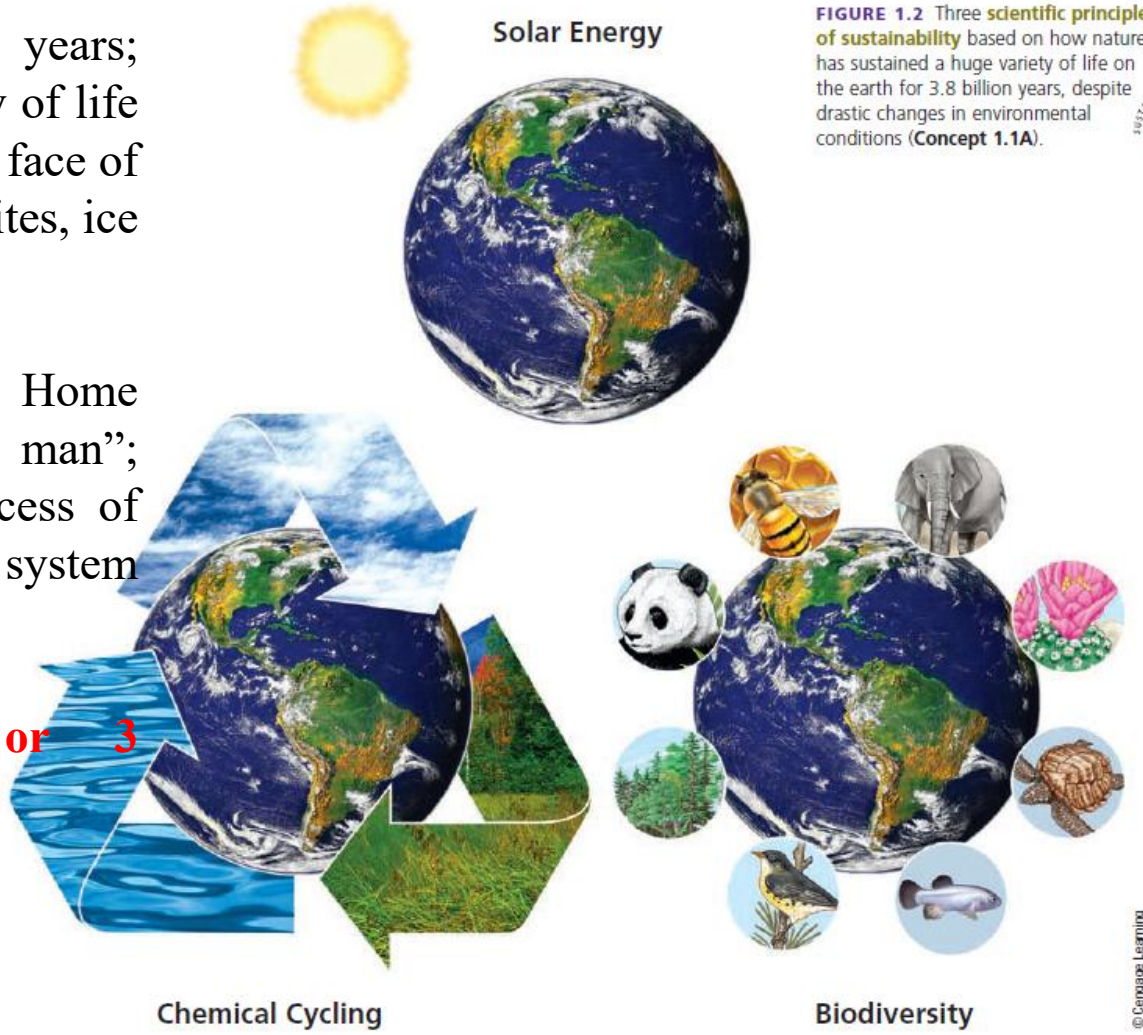


FIGURE 1.2 Three **scientific principles of sustainability** based on how nature has sustained a huge variety of life on the earth for 3.8 billion years, despite drastic changes in environmental conditions (**Concept 1.1A**).

Key Principles of Sustainability

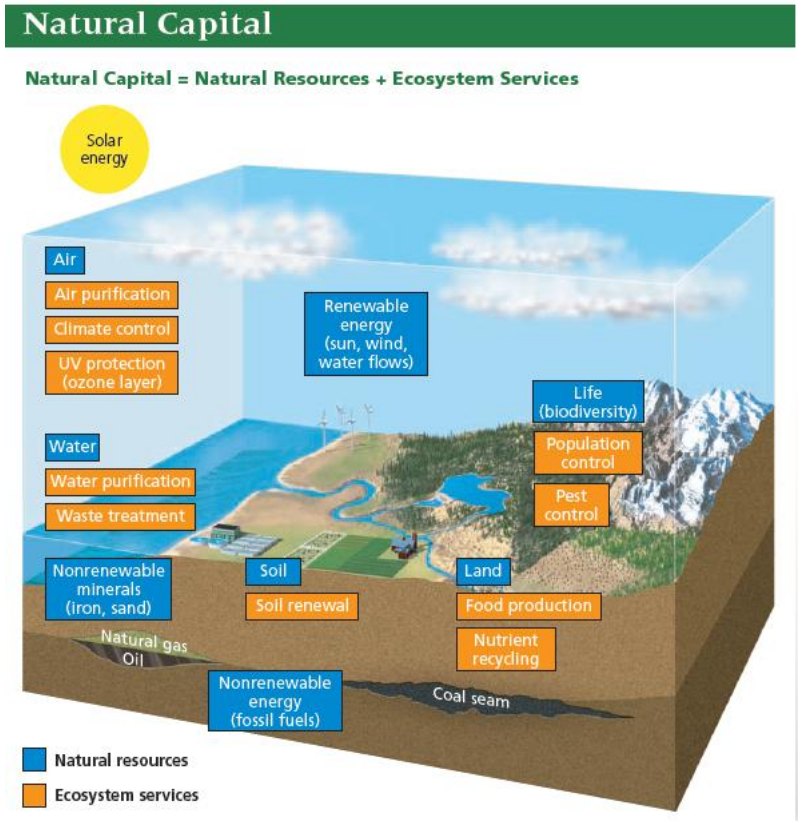
The clearing of vast areas of forest is an example of natural capital degradation.

1. Natural capital = Natural Resources + Natural Services; Natural Resources = Renewable + Nonrenewable Resources

2. To recognize human activities can degrade natural capital

3. Solutions

4.Trade-offs or compromises



Sustainable development



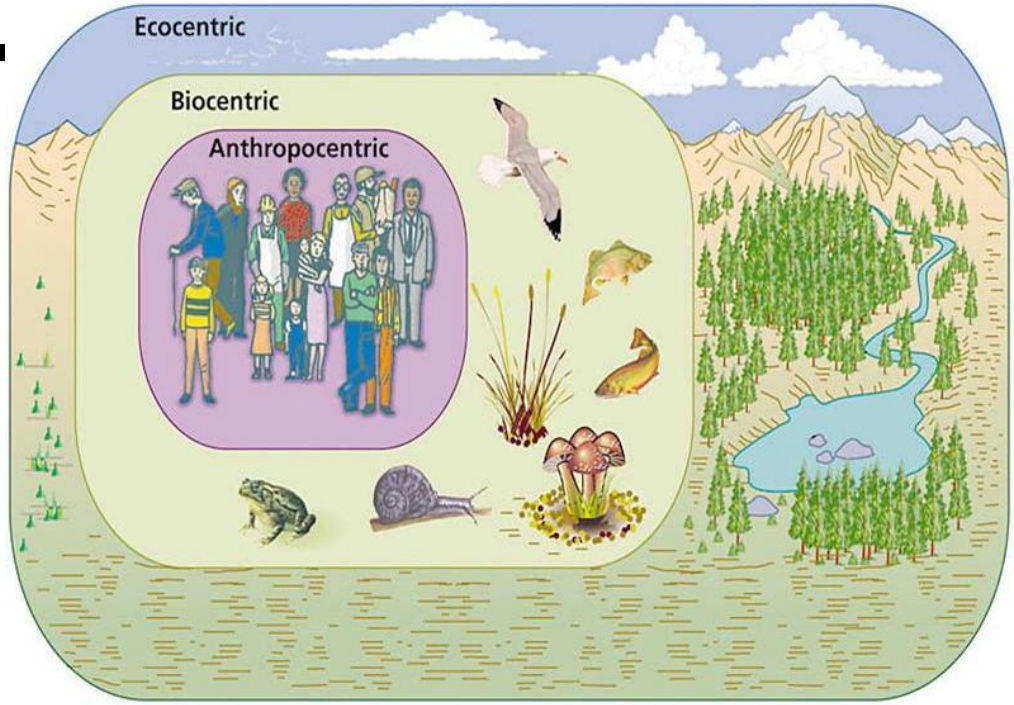
Other Principles of Sustainability

From Social Sciences



Environmental Ethics

"Which reconstructs the essential types of argumentation that can be made for protecting natural entities and sustainable use of natural resources"



Anthropocentrism: only humans have rights

- Costs and benefits are measured only according to their impact on people
- Anything not providing benefit to people has no value, humans are the center of universe.

Biocentrism: living things (plants, animals) have value

- All life has ethical standing
- Development is opposed if it destroys life, even if it creates jobs

Ecocentrism: includes abiotic factors such as rivers, estuaries, all ecological systems have value

- Values the well-being of species, communities, or ecosystems
- Holistic perspective, stresses preserving connections

Natural resources, are they infinite?



Inexhaustible
Solar energy
Wind energy
Geothermal energy



Renewable
Trees
Topsoil
Freshwater

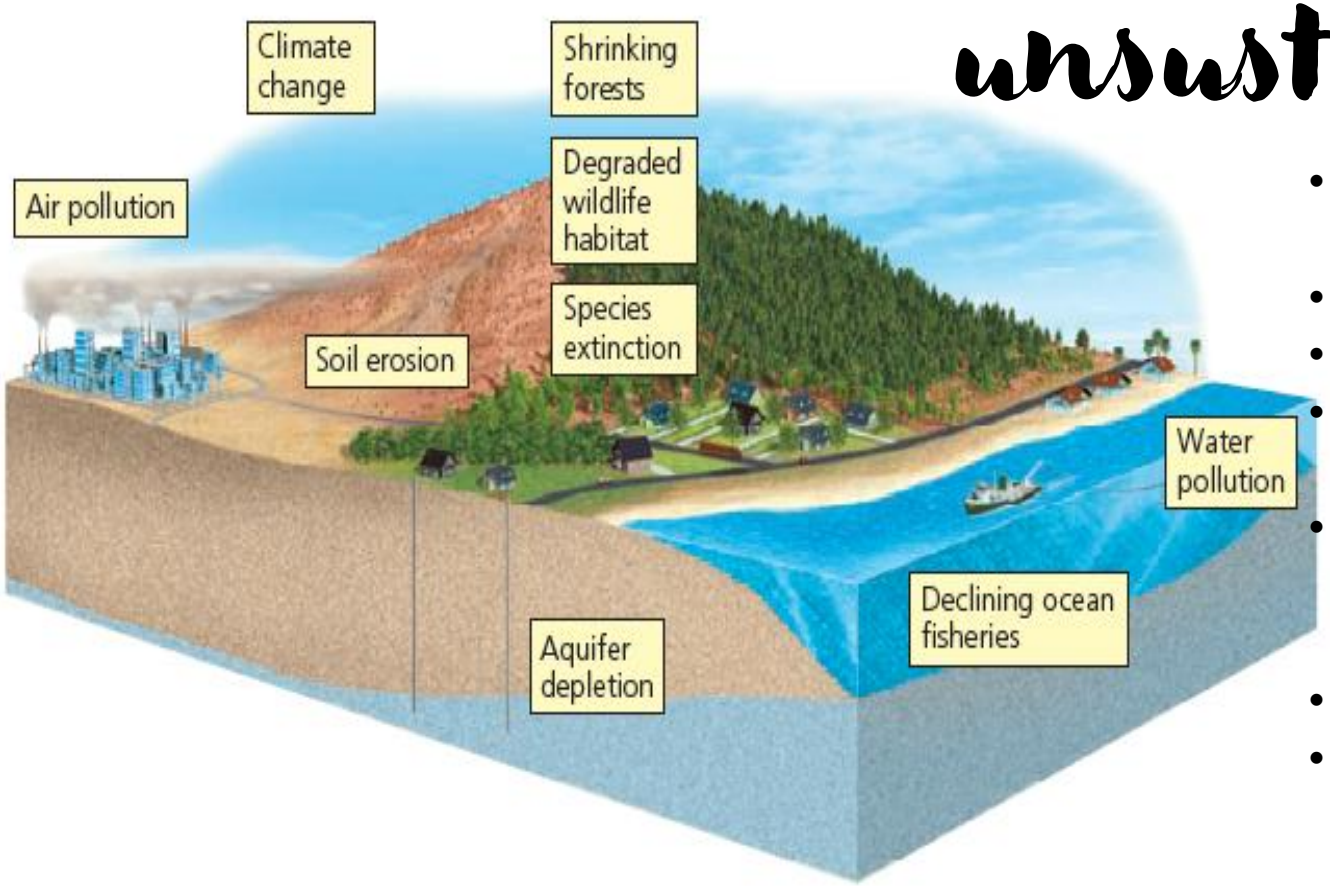


Nonrenewable (Exhaustible)
Fossil fuels (oil, natural gas, coal)
Iron and copper

Sustainable yield: The highest rate at which we can use a renewable resource indefinitely without reducing its available supply.

Natural Capital Degradation

Degradation of Normally Renewable Natural Resources



We are living unsustainably!!!

- Renewable forests are shrinking
- Deserts are expanding
- Topsoil is eroding
- Lower atmosphere is warming
- Floating ice and some glaciers are melting at unexpected rates
- Sea levels are rising
- Ocean acidity is increasing

- Floods, droughts, severe weather, and forest fires are more frequent in some areas
- Rivers are running dry
- Harvests of many species of fish are dropping sharply, and coral reefs are dying
- Species are becoming extinct at least 100 times faster than in pre-human times

The Tragedy of the Commons!

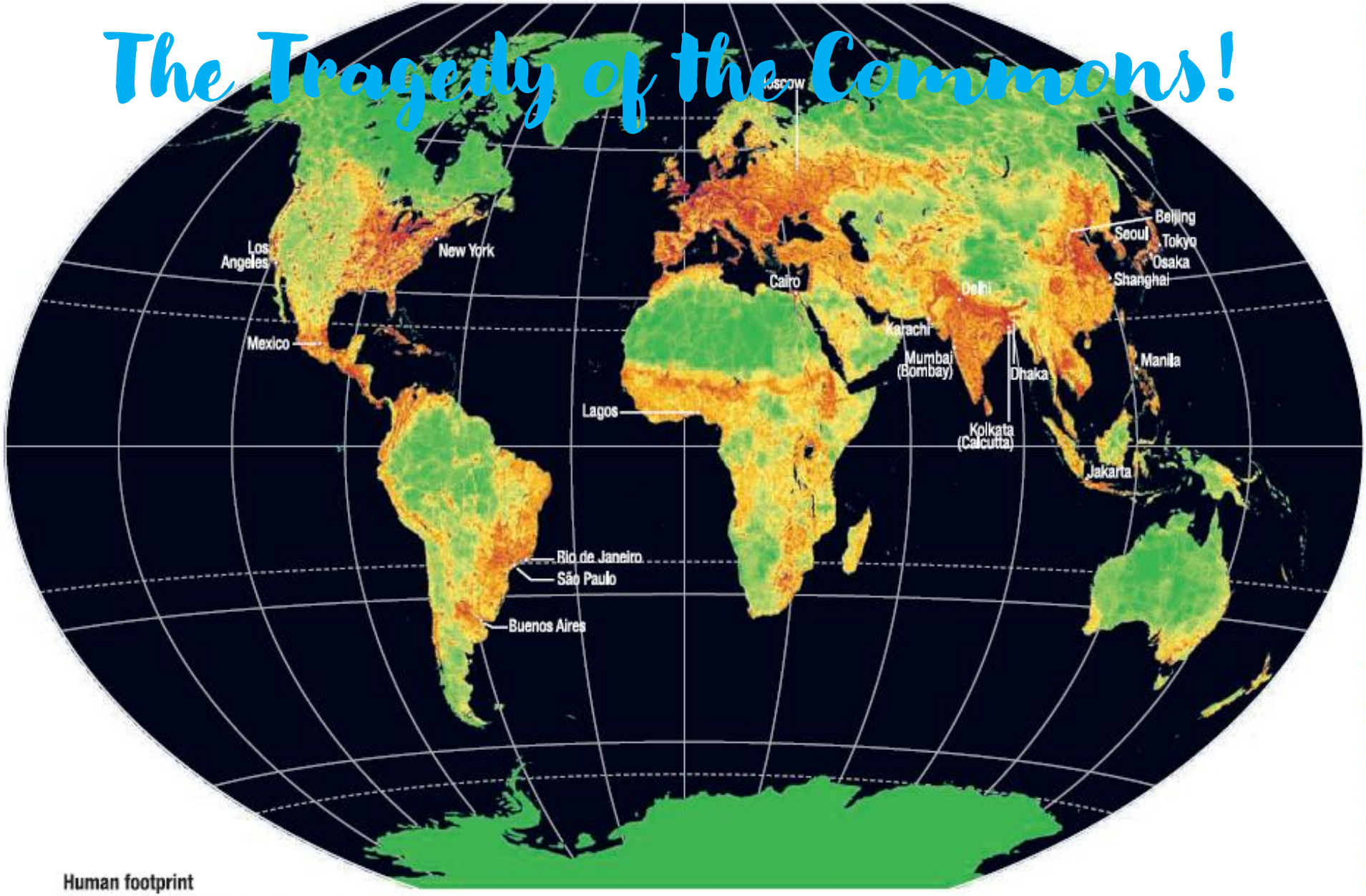


FIGURE 1.10 Natural capital use and degradation: The human ecological footprint has an impact on about 83% of the earth's total land surface.

Environmental Impact

Countries Differ in Resource Use and Environmental Impact



USA Vs India

TABLE 1.1 Environmental Impact of Selected High-, Middle-, and Low-Income Countries

Country	Population Size	Population Growth Rate	Resource Use Per Person	Use of Harmful Technology	Use of Beneficial Technology	Overall Environmental Impact
High-Income Countries						
United States	316 million	Moderate (0.5%)	Very high	Moderate	High	High
Japan	128 million	Negative (−0.2%)	High	Moderate	High	Moderate
Germany	82 million	Negative (−0.2%)	High	Moderate	High	Moderate
Middle-Income Countries						
China	1.35 billion	Moderate (0.5%)	Low	High	Moderate	High
India	1.26 billion	High (1.5%)	Low	High	Low	High
Brazil	194 million	Moderate (1%)	Low	High	Moderate	Moderate
Low-Income Countries						
Nigeria	402 million	High (2.6%)	Very low	High	Low	Moderate
Bangladesh	228 million	High (1.6%)	Very low	High	Low	Moderate
Congo	194 million	High (2.8%)	Very low	High	Low	Moderate

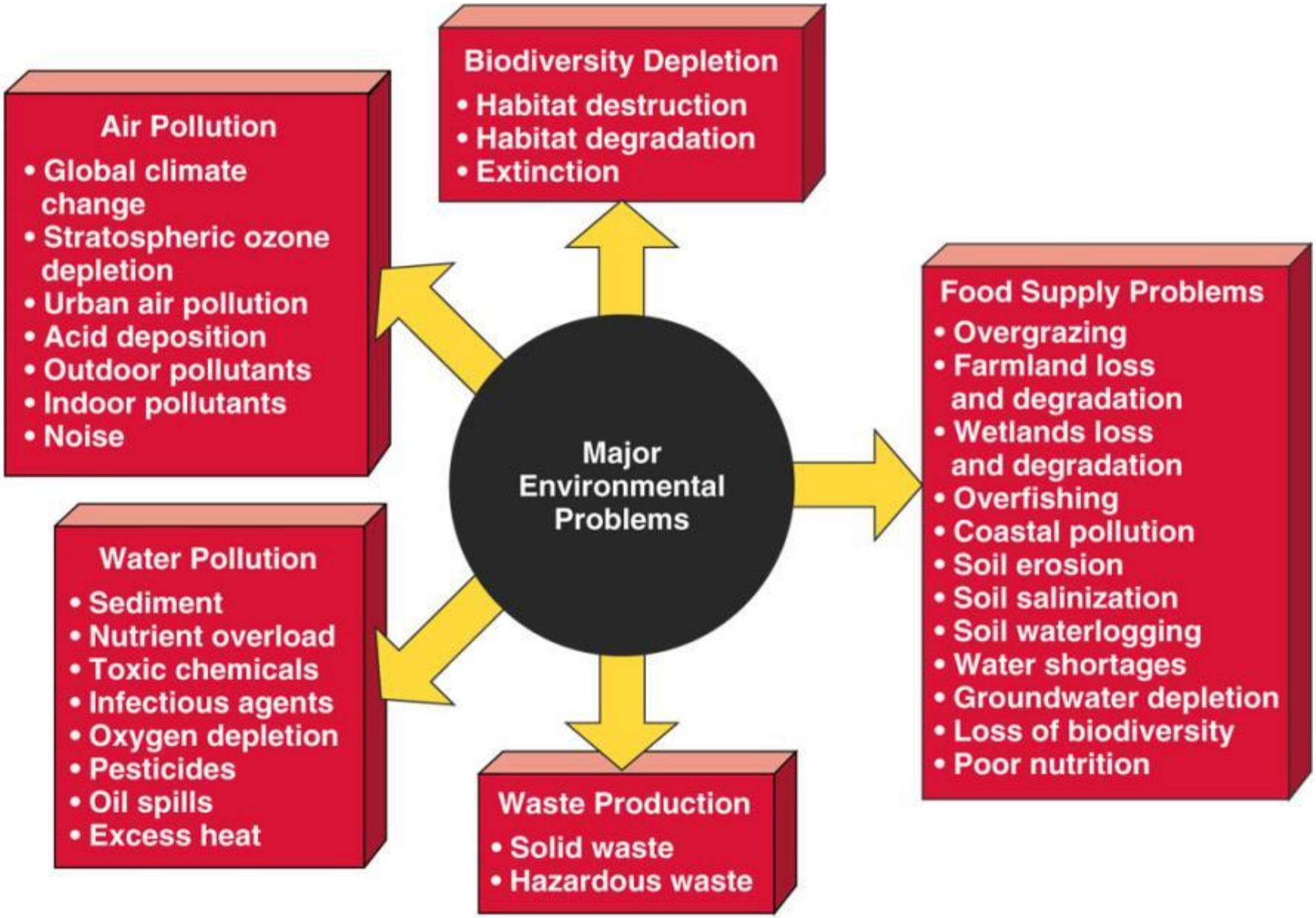
Compiled by the authors using data from Population Reference Bureau, Global Footprint Network, World Wide Fund for Nature, and Earth Policy Institute.

Copenhagen: A beautiful and most sustainable city





Major Environmental Problems



Major Environmental Problems



Major Environmental Problems

1. **Pollution: Air, water, soil Pollution:**

- Air Pollution: Industry and motor vehicle exhaust are prime causes. Air pollution is also caused by various gases and toxins released by industries and factories and combustion of fossil fuels.
- Soil Pollution: Heavy metals, nitrates and plastic are toxins which deplete the nutrients from soil.
- Water pollution: oil spills, acid rain, urban runoffs.



Environmental Problems (contd.)

2. Global Warming:

- Result of human practices causing emission of Greenhouse gases.

- Leads to rising temperatures of the oceans and the earth's surface causing melting of polar ice caps, rise in sea levels and also unnatural patterns of precipitation such as flash floods, excessive snow or desertification.



Major Environmental Problems (contd.)

3. Overpopulation:

- The population of the planet is reaching unsustainable levels as it faces shortage of resources like water, fuel and food.
- Population explosion in less developed and developing countries is straining the already scarce resources.
- Intensive agriculture practiced to produce food damages the environment through use of chemical fertilizer, pesticides and insecticides.
- Overpopulation is one of the crucial current environmental problem.



Major Environmental Problems (contd.)

4. Natural Resource Depletion:

- Another crucial current environmental problems.
- Fossil fuel consumption results in emission of Greenhouse gases, which is responsible for global warming and climate change.
- Globally, people are taking efforts to shift to renewable sources of energy like solar, wind, biogas and geothermal energy.
- The cost of installing the infrastructure and maintaining these sources has plummeted in the recent years.





Major Environmental Problems (contd.)

5. Waste Disposal:

- The over consumption of resources and creation of plastics are creating a global crisis of waste disposal.
- Nuclear waste disposal has tremendous health hazards associated with it.
- Plastic, fast food, packaging and cheap electronic wastes threaten the well-being of humans.
- Waste disposal is one of urgent current environmental problem.





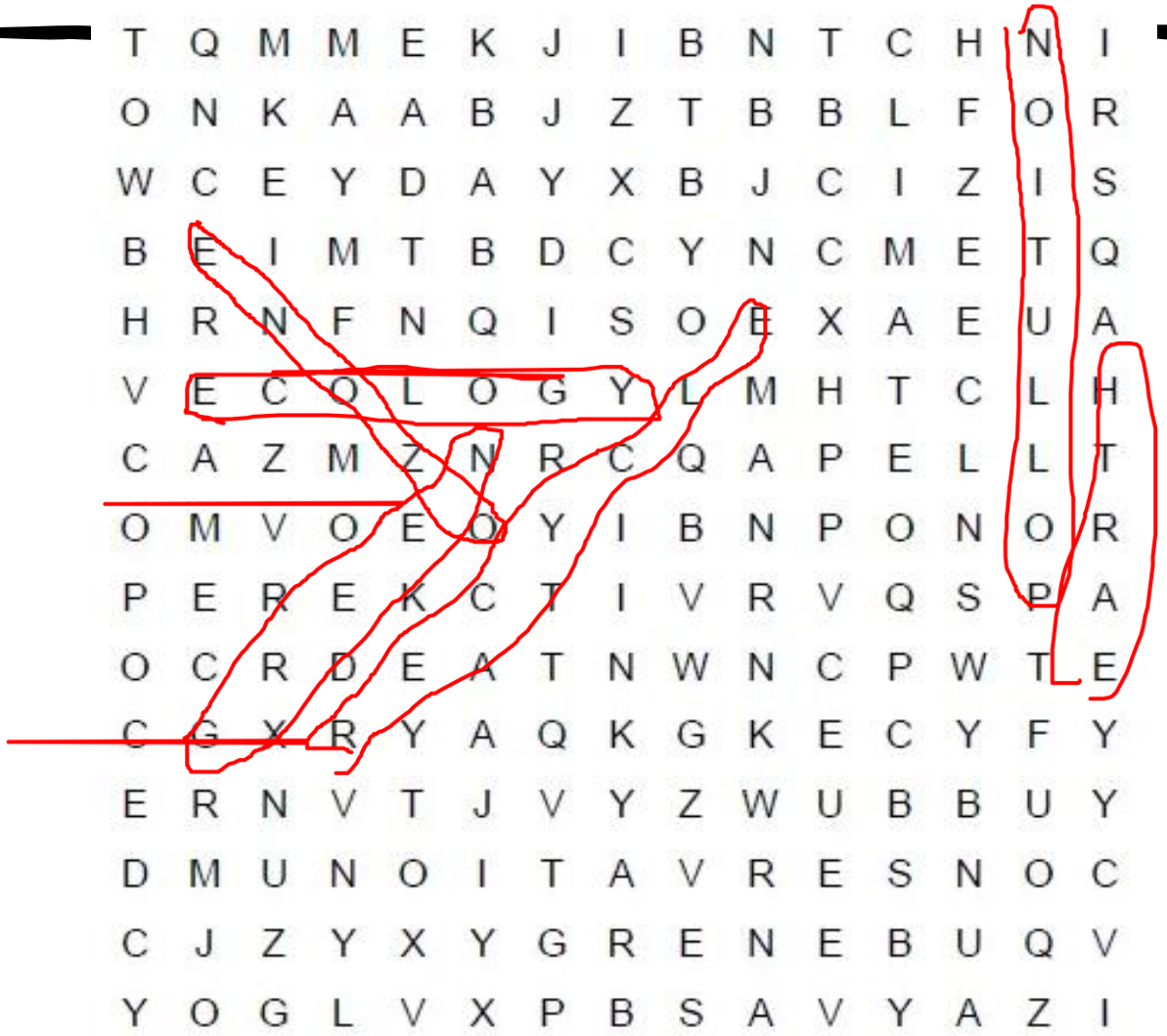


Word Search

CLIMATE
DAY
ENERGY
HABITAT
RECYCLE

COMPOST
EARTH
ENVIRONMENT
OZONE

CONSERVATION
ECOLOGY
GREEN
POLLUTION



Major Environmental Problems (contd.)

6. Climate Change:

- Another environmental problem that has surfaced in last couple of decades.
- It occurs due to rise in global warming which occurs due to increase in temperature of atmosphere by burning of fossil fuels and release of harmful gases by industries.
- Climate change has various harmful effects but not limited to melting of polar ice, change in seasons, occurrence of new diseases, frequent occurrence of floods and change in overall weather scenario.

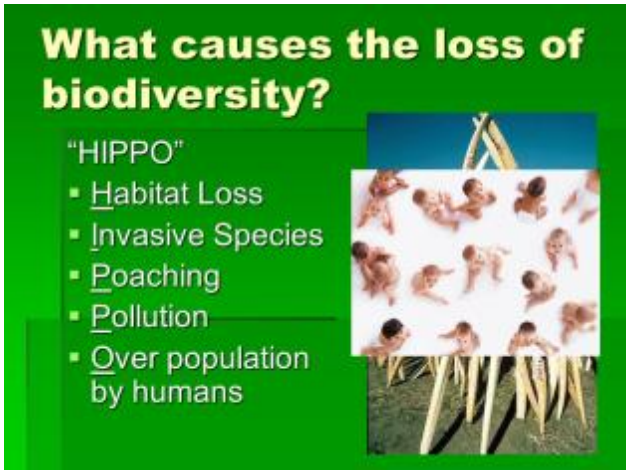




Major Environmental Problems (contd.)

7. Loss of Biodiversity:

- Human activity is leading to the extinction of species and habitats and loss of bio-diversity.
- Ecosystems, which took millions of years to perfect, are in danger when any species population is decimating.
- Balance of natural processes like pollination is crucial to the survival of the eco-system and human activity threatens the same.
- Another example is the destruction of coral reefs in the various oceans, which support the rich marine life





Major Environmental Problems (contd.)

8. **Deforestation:**

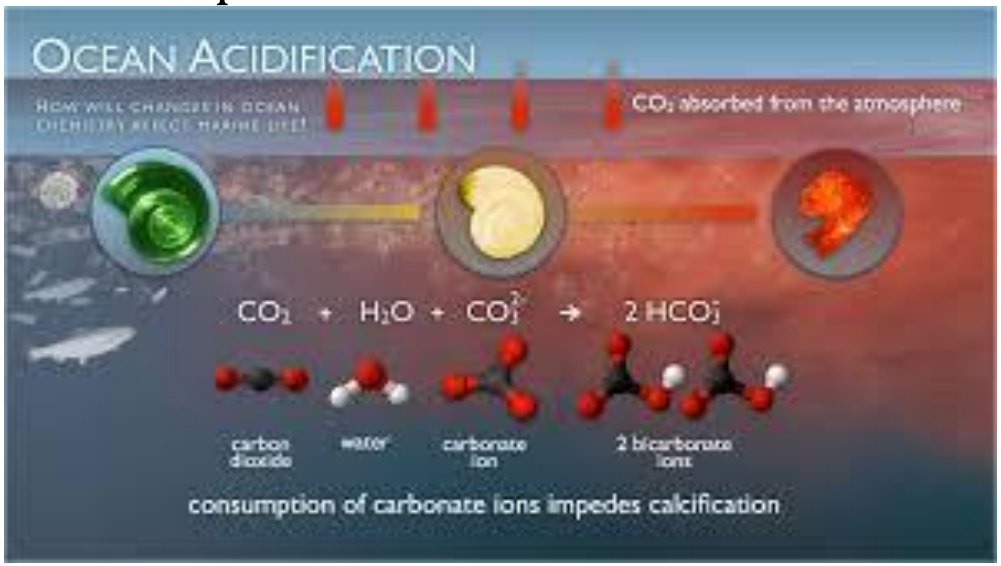
- Our forests are natural sinks of carbon dioxide and produce fresh oxygen as well as helps in regulating temperature and rainfall.
- At present, forests cover 30% of the land but every year tree cover is lost amounting to the country of Panama due to growing population demand for more food, shelter and cloth.
- Deforestation simply means clearing of green cover and make that land available for residential, industrial or commercial purpose.



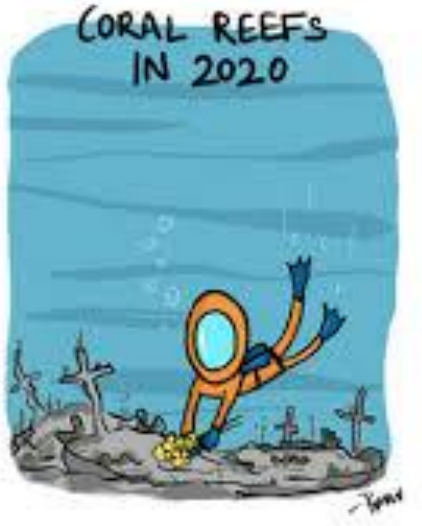
Major Environmental Problems (contd.)

9. Ocean Acidification:

- It is a direct impact of excessive production of CO_2. 25% of CO_2 produced by humans.
- The ocean acidity has increased by 30% in the last 250 years but by 2100, it may shoot up by 150%.
- The main impact is on shellfish and plankton in the same way as human osteoporosis.



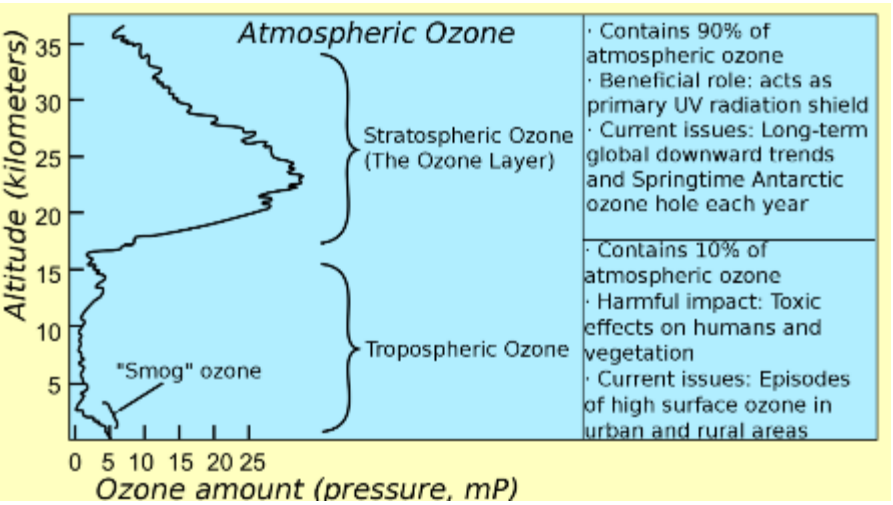


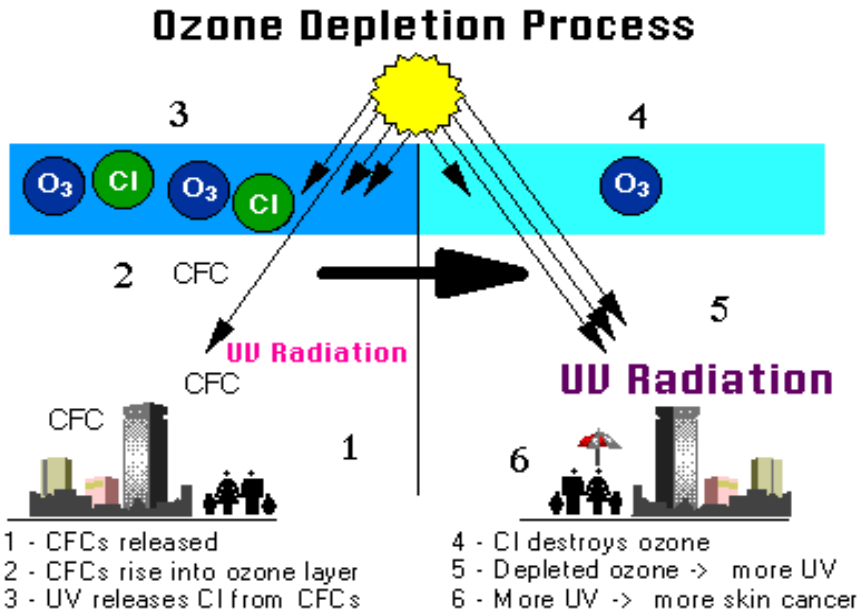


Major Environmental Problems (contd.)

10. Ozone Layer Depletion:

- The ozone layer is an invisible layer of protection around the planet that protects us from the sun's harmful rays.
- Depletion of ozone layer is caused by chlorine (chloroflorocarbons (CFC's)) and bromide (halons and methyl bromide). Once these toxic gases reach the upper atmosphere, they cause a hole in the ozone layer, the biggest of which is above the Antarctic.

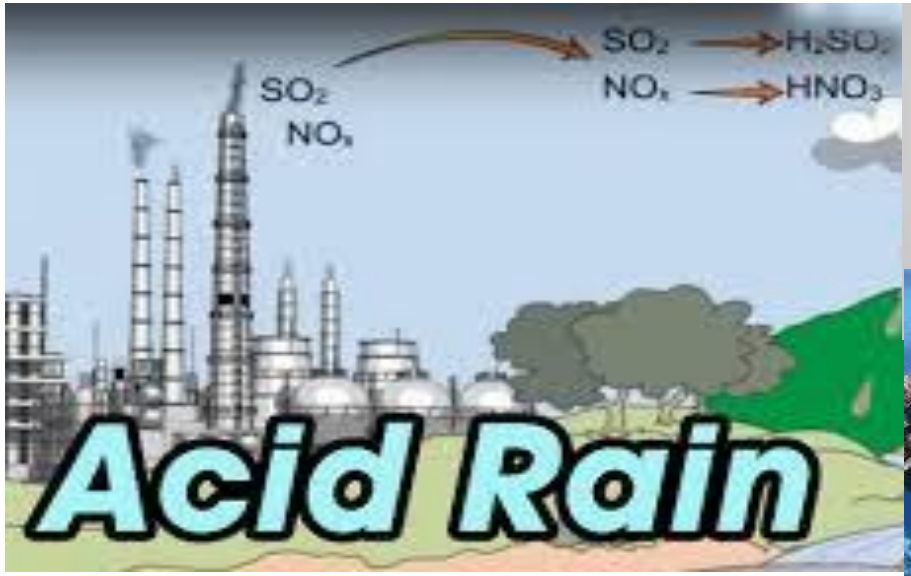




Major Environmental Problems (contd.)

11. Acid Rain:

- Acid rain occurs due to the presence of certain pollutants in the atmosphere.
- Acid rain can be caused due to combustion of fossil fuels or erupting volcanoes or rotting vegetation which release sulfur dioxide and nitrogen oxides into the atmosphere.
- Acid rain is a known environmental problem that can have serious effect on human health, wildlife and aquatic species.











Major Environmental Problems (contd.)

12. Urban Sprawl:

- Urban sprawl refers to migration of population from high density urban areas to low density rural areas which results in spreading of city over more and more rural land.
- Urban sprawl results in land degradation, increased traffic, environmental issues and health issues.
- The ever growing demand of land displaces natural environment consisting of flora and fauna instead of being replaced.



Major Environmental Problems (contd.)

13. Public Health Issues:

- The current environmental problems pose a lot of risk to health of humans, and animals.
- Dirty water is the biggest health risk of the world and poses threat to the quality of life and public health.
- Run-off to rivers carries along toxins, chemicals and disease carrying organisms.
- Pollutants cause respiratory diseases like Asthma and cardiac-vascular problems. High temperatures encourage the spread of infectious diseases like Dengue.





Environmental Problems: Causes

- Consumerism: Consumption of natural resources by the people; wasteful and unsustainable resource use



constant purchasing



More technological development, more consumption of natural resources

- Population, more consumption; large waste generation



- Failure to include the harmful environmental and health costs of goods and services in their market prices
- Increasing isolation from nature
- Poverty
- Lifestyle





Consumerism


1 NO POVERTY
2 ZERO HUNGER
3 GOOD HEALTH AND WELL-BEING
4 QUALITY EDUCATION
5 GENDER EQUALITY
6 CLEAN WATER AND SANITATION
7 AFFORDABLE AND CLEAN ENERGY
8 DECENT WORK AND ECONOMIC GROWTH
9 INDUSTRY, INNOVATION AND INFRASTRUCTURE
10 REDUCED INEQUALITIES
11 SUSTAINABLE CITIES AND COMMUNITIES
THE GLOBAL GOALS
For Sustainable Development
12 RESPONSIBLE CONSUMPTION AND PRODUCTION
13 CLIMATE ACTION
14 LIFE BELOW WATER
15 LIFE ON LAND
16 PEACE AND JUSTICE STRONG INSTITUTIONS
17 PARTNERSHIPS FOR THE GOALS

How to save our planet??

Tutorial 1

1. Why should we care about the environment?

2. Are we the most important beings on the planet? Describe your answer in a small paragraph.

3. Do we have an ethical obligation to pass on to future generations the extraordinary natural world in a condition that is at least as good as what we inherited? What do you think?


Thank you!